

CMP-27 - Human Interaction Layer

Own Interaction Sessions, modality coordination, accessibility, consent cues, clarification, confirmation, and human-visible action status.

Authority and state ownership

Book I: Decisions 1-7, 24-29, 35, 40, 47-50, and 58. Book II: Chapters 14, 31. Authoritative state: Interaction Sessions, turn and modality state, clarification and confirmation records, accessibility preferences. Explicitly non-owned: Presence Objects, human Identity Records, Intent, Authority, action execution.

Critical invariant

Human-facing language, convenience, or conversational assent never substitutes for governed identity, Authority, or the exact protected Permission decision.

Logical interfaces

CMP-27-IF-01 / Command / Start Interaction

Command `Start Interaction` MUST carry identity, authority context where applicable, schema version, correlation, causation, time/freshness, provenance, and explicit success or denial semantics. Book IX owns wire details.

CMP-27-IF-02 / Command / Submit Human Input

Command `Submit Human Input` MUST carry identity, authority context where applicable, schema version, correlation, causation, time/freshness, provenance, and explicit success or denial semantics. Book IX owns wire details.

CMP-27-IF-03 / Command / Request Confirmation

Command `Request Confirmation` MUST carry identity, authority context where applicable, schema version, correlation, causation, time/freshness, provenance, and explicit success or denial semantics. Book IX owns wire details.

CMP-27-IF-04 / Command / End Interaction

Command `End Interaction` MUST carry identity, authority context where applicable, schema version, correlation, causation, time/freshness, provenance, and explicit success or denial semantics. Book IX owns wire details.

CMP-27-IF-05 / Query / Get Session State

Query `Get Session State` MUST carry identity, authority context where applicable, schema version, correlation, causation, time/freshness, provenance, and explicit success or denial semantics. Book IX owns wire details.

CMP-27-IF-06 / Query / Get Accessible Representation

Query `Get Accessible Representation` MUST carry identity, authority context where applicable, schema version, correlation, causation, time/freshness, provenance, and explicit success or denial semantics. Book IX owns wire details.

CMP-27-IF-07 / Query / Get Action Status

Query `Get Action Status` MUST carry identity, authority context where applicable, schema version, correlation, causation, time/freshness, provenance, and explicit success or denial semantics. Book IX owns wire details.

CMP-27-IF-08 / Event / Interaction Started

Event `Interaction Started` MUST carry identity, authority context where applicable, schema version, correlation, causation, time/freshness, provenance, and explicit success or denial semantics. Book IX owns wire details.

CMP-27-IF-09 / Event / Clarification Requested

Event `Clarification Requested` MUST carry identity, authority context where applicable, schema version, correlation, causation, time/freshness, provenance, and explicit success or denial semantics. Book IX owns wire details.

CMP-27-IF-10 / Event / Confirmation Recorded

Event `Confirmation Recorded` MUST carry identity, authority context where applicable, schema version, correlation, causation, time/freshness, provenance, and explicit success or denial semantics. Book IX owns wire details.

CMP-27-IF-11 / Event / Interaction Ended

Event `Interaction Ended` MUST carry identity, authority context where applicable, schema version, correlation, causation, time/freshness, provenance, and explicit success or denial semantics. Book IX owns wire details.

Lifecycle, controls, and recovery

Lifecycle: Proposed > Active > Awaiting Clarification or Confirmation > Completed or Abandoned. All transitions require version, guard, identity, authority where applicable, idempotency, time context, and evidence. Contain: identity confusion, dark pattern, inaccessible output, consent ambiguity, stale action status. Recovery proves identity, state, authority, provenance, dependencies, and invariants before mutation resumes.

Conformance tests

CMP-27-TST-001 - Sole-owner mutation. Attempt mutation through a consumer, replica, cache, operator path, and direct datastore route. Expected: Every unauthorized path is denied and evidenced.

CMP-27-TST-002 - Authority denial. Invoke every command with absent, expired, revoked, mismatched, and over-broad authority context. Expected: No state changes; denial reason and evidence are complete.

CMP-27-TST-003 - Valid lifecycle. Exercise every declared transition with valid guards and fresh versions. Expected: One transition Evidence Object is admitted per transition identifier; duplicate attempts produce no additional state mutation or admitted Evidence Object.

CMP-27-TST-004 - Invalid lifecycle. Attempt skipped, stale, replayed, and terminal-state transitions. Expected: Every invalid transition is rejected without partial mutation.

CMP-27-TST-005 - Critical invariant. Human-facing language, convenience, or conversational assent never substitutes for governed identity, Authority, or the exact protected Permission decision. Expected: Invariant holds under normal, concurrent, degraded, and recovery conditions.

CMP-27-TST-006 - Failure containment. Inject identity confusion, dark pattern, inaccessible output, consent ambiguity, stale action status. Expected: Unsafe work stops, evidence is preserved, and declared containment state is observable.

CMP-27-TST-007 - Recovery. Restore from a verified checkpoint, reconcile dependencies, and resume after validation. Expected: Identity, state, authority, provenance, and evidence continuity are proven.

CMP-27-TST-008 - Contract compatibility. Run current and prior supported contract versions plus an unsupported version. Expected: Supported behavior remains compatible; unsupported input fails explicitly.

CMP-27-TST-009 - Privacy and security. Inject secrets, personal data, malicious input, confused-deputy requests, and log-exfiltration probes. Expected: Least privilege and data controls hold; prohibited data is absent from telemetry.

CMP-27-TST-010 - Topology independence. Repeat critical tests under permitted single-node, replicated, partitioned, and restored topologies. Expected: Topology does not multiply authority or mutation ownership.

Traceability and certification

Twelve requirements map to Books I-III and Book X. No open finding or Owner Review item. Final for Book IV v1.0.